

# Smaller 5-chromatic unit-distance graphs on two icosahedral spheres

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## Abstract

Voronov, Neopryatnaya and Dergachev constructed 5-chromatic unit-distance graphs on the sphere  $S^2(r_1)$ ,  $r_1 = \cos \frac{\pi}{10}$ , with 372 vertices, and on  $S^2(r_2)$ ,  $r_2 = \cos \frac{3\pi}{10}$ , with 972 vertices, and noted that the first graph stays 5-chromatic after deleting any single vertex. By SAT-based vertex minimization we find a 5-chromatic subgraph of the first graph with 231 vertices and 938 edges, and one of the second with 961 vertices and 4028 edges. Both are vertex-critical and contain no Moser spindle; the smaller one is the best found by our search, not a proved minimum. All coordinates are given exactly in nested radicals and verified in exact arithmetic; non-4-colourability is certified by DRAT proofs checked with DRAT-TRIM.

## 1 Introduction

The chromatic number  $\chi(X)$  of a subset  $X \subseteq \mathbb{R}^n$  is the least number of colours needed to colour  $X$  so that no two points at Euclidean distance 1 receive the same colour. For the plane, de Grey [3] proved  $\chi(\mathbb{R}^2) \geq 5$ ; the smallest known 5-chromatic unit-distance graph in the plane has 509 vertices [7] (see also [4]).

For the sphere  $S^2(r) \subset \mathbb{R}^3$  of radius  $r > \frac{1}{2}$ , Simmons [8] studied colourings, and Voronov, Neopryatnaya and Dergachev [10] constructed the first 5-chromatic unit distance graphs embedded in spheres,  $G_{372} \subset S^2(r_1)$  and  $G_{972} \subset S^2(r_2)$ , where  $r_1 = \cos \frac{\pi}{10}$  and  $r_2 = \cos \frac{3\pi}{10}$ , proving  $\chi(S^2(r_1)) \geq 5$  and  $\chi(S^2(r_2)) \geq 5$ . They remark that  $\chi(G_{372}) = 5$  is preserved after deleting any single vertex and that they “do not give here the subgraphs of  $G_{372}$  that have the property of minimality” [10, Sect. 4]. Cherkashin and Voronov [2] ask, more generally, for the minimal number of vertices of a subgraph  $G$  of  $S^2(r)$  with  $\chi(G) = \chi(S^2(r))$ . We reduce both constructions.

**Theorem 1.** (a) *The graph  $H_{231} \subset G_{372}$  defined in Section 3 is a unit distance graph in  $S^2(r_1)$  with 231 vertices and 938 edges, and  $\chi(H_{231}) = 5$ .*

(b) *The graph  $H_{961} \subset G_{972}$  defined in Section 3 is a unit distance graph in  $S^2(r_2)$  with 961 vertices and 4028 edges, and  $\chi(H_{961}) = 5$ .*

*Both graphs are vertex-critical, i.e. deleting any vertex leaves a 4-colourable graph, and neither contains the Moser spindle as a subgraph.*

Thus  $H_{231}$  has 38% fewer vertices than  $G_{372}$ , while  $G_{972}$  turns out to be nearly vertex-critical (Remark 3). We do not know whether 231 is the least order of a 5-chromatic subgraph of  $G_{372}$ ; it is the smallest our search found. Since  $G_{372}$  and  $G_{972}$  are defined exactly in [10], our graphs are specified exactly by listing which of their vertices are kept or removed (Section 3 and Appendix A).

## 2 The parent graphs

Let  $\Gamma$  be the full symmetry group of the regular icosahedron (order 120, including reflections), generated as in [9] by

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \frac{1}{2} \begin{pmatrix} 1 & -\tau & \tau^{-1} \\ \tau & \tau^{-1} & -1 \\ \tau^{-1} & 1 & \tau \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \tau = \frac{1+\sqrt{5}}{2}.$$

$G_{372}$ . Its vertices are the 12 vertices of the icosahedron with unit edge (the  $\Gamma$ -orbit of  $(\frac{\tau}{2}, \frac{1}{2}, 0)$ ) together with the  $\Gamma$ -orbits of the three points  $v_1, v_2, v_3 \in S^2(r_1)$  of Table 1, each orbit of size 120; edges join vertices at distance 1. It has 372 vertices and 1710 edges [10].

|            | value                    | minimal polynomial of twice the coordinate               |
|------------|--------------------------|----------------------------------------------------------|
| $x_1$      | 0.71584...               | $x^4 - 2x^3 + 2x^2 - x - 1$                              |
| $y_1$      | 0.34924...               | $x^4 + x^3 - 2x^2 + 2x - 1$                              |
| $z_1$      | 0.51972...               | $x^8 - 5x^6 + 2x^4 + 20x^2 - 19$                         |
| $x_2, x_3$ | -0.07961..., -0.17646... | $x^8 - x^7 - 4x^6 + 6x^5 + 28x^4 - 7x^3 + 9x^2 + 8x + 1$ |
| $y_2, y_3$ | -0.08345..., 0.79929...  | $x^8 - 2x^7 + x^6 - 5x^5 + 9x^4 - 4x^3 + 5x^2 - 5x - 1$  |
| $z_2$      | 0.94404...               | $x^8 - 3x^7 + 10x^5 + 4x^4 - 24x^3 - 13x^2 + 7x - 1$     |
| $z_3$      | 0.48426...               | $x^8 + 3x^7 - 10x^5 + 4x^4 + 24x^3 - 13x^2 - 7x - 1$     |

Table 1: Orbit representatives  $v_i = (x_i, y_i, z_i)$  of  $G_{372}$ , from [10].

$G_{972}$ . Its vertices are the 12 vertices of the great icosahedron with unit edge (the  $\Gamma'$ -orbit of  $\tau^{-1}(\frac{1}{2}, \frac{\tau}{2}, 0)$ , where  $\Gamma'$  is  $\Gamma$  conjugated by the coordinate swap  $y \leftrightarrow z$ , as in [9]) together with the  $\Gamma'$ -orbits of eight points  $w_1, \dots, w_8 \in S^2(r_2)$ , each of size 120. The coordinates of  $w_i$  are given in [9] as roots of explicit polynomials of degree 8 (reproduced in the repository, `construction/build.G972.py`);

for instance the second coordinate of  $w_8$  is  $\frac{1}{4}(1 - \sqrt{6 - 3\sqrt{5}} + 2\sqrt{(17\sqrt{5} - 37)/2})$ . The graph has 972 vertices and 4110 edges.

**Lemma 2.** *For each sphere, all coordinates of all vertices of the parent graph lie in a number field  $K$  of degree 8 whose Galois closure has group  $D_8$  (of order 16). Consequently  $K$  is a tower of quadratic extensions  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{5}) \subset \mathbb{Q}(\sqrt{5}, b) \subset \mathbb{Q}(\sqrt{5}, b, g) = K$  with  $b = \sqrt{\beta}$ ,  $g = \sqrt{\gamma}$ , where*

$$\begin{aligned} S^2(r_1): \quad \beta &= \frac{7}{2} + \frac{5}{2}\sqrt{5}, & \gamma &= \frac{15}{4} - \frac{7}{4}\sqrt{5} + \left(\frac{3}{4} - \frac{1}{4}\sqrt{5}\right)b, \\ S^2(r_2): \quad \beta &= \frac{23}{2} + \frac{11}{2}\sqrt{5}, & \gamma &= \frac{5}{4} - \frac{3}{4}\sqrt{5} + \left(-\frac{1}{4} + \frac{1}{4}\sqrt{5}\right)b, \end{aligned}$$

*all square roots being positive. Every coordinate of every vertex is a  $\mathbb{Q}$ -linear combination of  $1, \sqrt{5}, b, \sqrt{5}b, g, \sqrt{5}g, bg, \sqrt{5}bg$ , i.e. an explicit nested radical with rational coefficients. For example, vertex 13 of  $H_{231}$  (Appendix A) is*

$$\left(\frac{1}{4} + \frac{\sqrt{5}-1}{8}b, \quad -\frac{1+\sqrt{5}}{8} + \frac{1}{4}b, \quad \frac{1+\sqrt{5}}{4}g\right) \quad \text{on } S^2(r_1).$$

The field and the tower were computed with PARI/GP [6]; the exact coordinates of all vertices of  $H_{231}$  and  $H_{961}$  in this form, written with integers, fractions and square roots only, are in the repository (Mathematica and Python syntax).

### 3 The graphs $H_{231}$ and $H_{961}$

The exact coordinates of every vertex are given as nested radicals (Lemma 2) in the repository. For readability, the tables below list coordinates rounded to four decimals; since the vertices of  $G_{372}$  are pairwise at distance at least 0.049, and those of  $G_{972}$  at least 0.009, these rounded triples identify the exact vertices unambiguously. Exact radical forms of all orbit representatives are given in Appendix B, so every vertex is defined exactly within this paper.

$H_{231}$  is the subgraph of  $G_{372}$  induced by the 231 vertices listed in Appendix A: all 12 icosahedron vertices, 99 vertices of the orbit of  $v_1$ , and 60 vertices of each of the orbits of  $v_2$  and  $v_3$ . Its symmetry group within  $\Gamma$  is trivial.

$H_{961}$  is obtained from  $G_{972}$  by deleting the 11 vertices of Table 2.

|    | orbit        | $x$     | $y$     | $z$     |
|----|--------------|---------|---------|---------|
| 1  | $w_3$ -orbit | -0.1495 | -0.4014 | 0.4025  |
| 2  | $w_4$ -orbit | 0.5257  | 0.1359  | -0.2250 |
| 3  | $w_4$ -orbit | -0.1650 | 0.5628  | -0.0388 |
| 4  | $w_4$ -orbit | 0.1228  | 0.3849  | 0.4269  |
| 5  | $w_4$ -orbit | -0.1599 | 0.2878  | 0.4869  |
| 6  | $w_4$ -orbit | 0.2250  | -0.5257 | -0.1359 |
| 7  | $w_4$ -orbit | 0.4869  | -0.1599 | 0.2878  |
| 8  | $w_4$ -orbit | 0.0600  | 0.4237  | -0.4029 |
| 9  | $w_4$ -orbit | -0.3849 | -0.4269 | -0.1228 |
| 10 | $w_4$ -orbit | 0.5628  | 0.0388  | 0.1650  |
| 11 | $w_4$ -orbit | -0.4237 | -0.4029 | -0.0600 |

Table 2: The 11 vertices of  $G_{972}$  deleted to obtain  $H_{961}$  (four-decimal labels).

Table 3 summarizes both graphs and Figure 1 shows  $H_{231}$ .

| graph          | sphere     | $ V $ | $ E $ | degrees          | composition by orbit     | vertex-critical |
|----------------|------------|-------|-------|------------------|--------------------------|-----------------|
| $G_{372}$ [10] | $S^2(r_1)$ | 372   | 1710  |                  | $12 + 120 + 120 + 120$   | no              |
| $H_{231}$      | $S^2(r_1)$ | 231   | 938   | 4–25 (avg. 8.12) | $12 + 99 + 60 + 60$      | yes             |
| $G_{972}$ [10] | $S^2(r_2)$ | 972   | 4110  |                  | $12 + 8 \times 120$      | no              |
| $H_{961}$      | $S^2(r_2)$ | 961   | 4028  | 5–65 (avg. 8.38) | $12 + 8 \times 120 - 11$ | yes             |

Table 3: The parent graphs and the reduced graphs.

### 4 Proof of Theorem 1

The proof has four parts; parts (ii)–(iv) are finite computations, each accompanied by a certificate that can be checked independently of the software that produced it.

**(i) Embedding and edges.** By Lemma 2 all vertices have coordinates in a field  $K$  of degree 8. In PARI/GP [6] we represent  $K$  as  $\mathbb{Q}[y]/(F(y))$ , compute each coordinate of every orbit representative as an element of  $K$  (the roots of the minimal polynomials in  $K$ , choosing the real embedding of

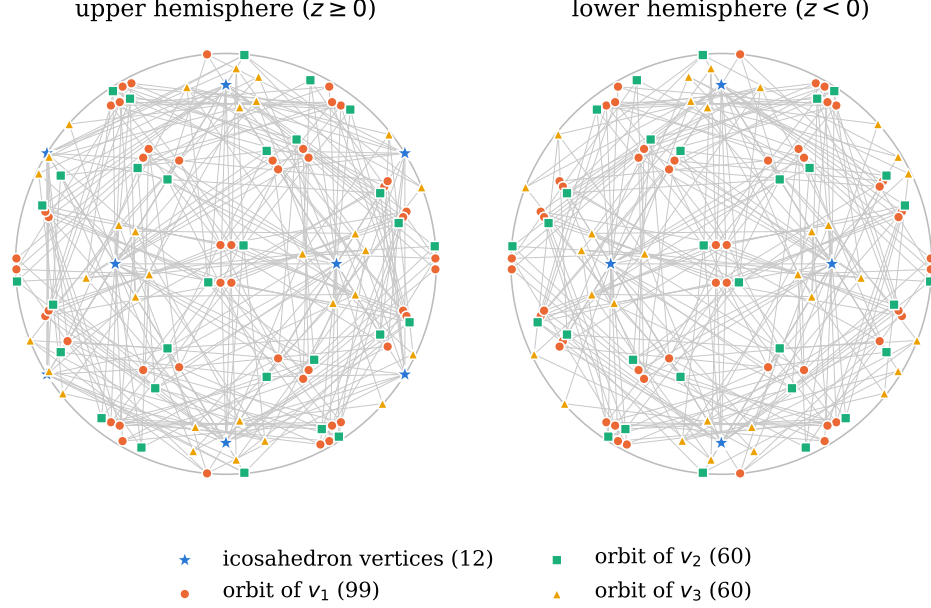


Figure 1: Orthogonal projections of  $H_{231}$  onto the  $xy$ -plane: vertices with  $z \geq 0$  (left) and  $z < 0$  (right, viewed from below). Marker shape and colour indicate the orbit of  $G_{372}$  a vertex belongs to; edges between vertices of the same hemisphere are drawn in grey. The bounding circle has radius  $r_1$ .

$K$  that matches the numerical values), generate the 120 matrices of  $\Gamma$  (resp.  $\Gamma'$ ) exactly, and form all vertices. For every vertex  $p$  of  $H_{231}$  and  $H_{961}$  we check  $p \cdot p = r^2$  exactly, with  $r_1^2 = \frac{5+\sqrt{5}}{8}$  and  $r_2^2 = \frac{5-\sqrt{5}}{8}$ , and for every edge  $pq$  we check  $|p - q|^2 = 1$  exactly (938 and 4028 identities in  $K$ ; all hold). Conversely, no unit distance is missing: with the coordinates to 30 significant digits (files `H231_vertices.txt`, `H961_vertices.txt`), every non-adjacent pair satisfies  $||p - q| - 1| \geq 2.2 \cdot 10^{-3}$  in  $H_{231}$  and  $\geq 4.1 \cdot 10^{-4}$  in  $H_{961}$ , far above the numerical error. Hence both graphs are unit-distance graphs on the stated spheres with the stated numbers of edges.

This was double-checked independently of PARI/GP: a short program implements exact arithmetic in the basis  $1, \sqrt{5}, b, \sqrt{5}b, g, \sqrt{5}g, bg, \sqrt{5}bg$  with rational coefficients (using only the relations  $\sqrt{5}^2 = 5, b^2 = \beta, g^2 = \gamma$ ), reads the nested-radical coordinates from the files `H231_exact_coordinates.m` and `H961_exact_coordinates.m`, and confirms  $p \cdot p = r^2$  for all vertices and  $|p - q|^2 = 1$  for all 938 and 4028 edges; as a negative control, perturbing a single coordinate by  $10^{-3}$  is detected (the vertex leaves the sphere and all its edges fail). The edges also form a subset of the edges of  $G_{372}$  and  $G_{972}$ , whose unit distances were verified symbolically in [10, 9].

**(ii) No proper 4-colouring.** We encode the existence of a proper 4-colouring as a CNF formula with variables  $x_{v,c}$ , clauses  $\bigvee_c x_{v,c}$  and  $\neg x_{v,c} \vee \neg x_{v,c'}$  ( $c \neq c'$ ) for every vertex  $v$ , clauses  $\neg x_{u,c} \vee \neg x_{v,c}$  for every edge  $uv$  and colour  $c$ , and two unit clauses fixing the colours of the end-points of one edge (which loses no generality). The SAT solver KISSAT [1] reports both formulas unsatisfiable and emits proofs in the DRAT format; the independent checker DRAT-TRIM [11] validates both proofs (s VERIFIED; files `H231_4col.cnf`, `H231_4col.drat.xz` and likewise for  $H_{961}$ ). As further independent confirmation, the solvers CaDiCaL, Glucose and MiniSat (via PySAT [5]) and a second encoding with two Boolean variables per vertex give the same answer

(`verification/crosscheck_solvers.py`), and for  $H_{231}$  a plain backtracking search (DSATUR vertex order, colour-symmetry breaking, about  $7 \cdot 10^6$  nodes, no SAT solver) finds no proper 4-colouring. Hence  $\chi \geq 5$ .

**(iii) A proper 5-colouring.** A 5-colouring of  $H_{231}$  is listed in Appendix A (colour classes of sizes 55, 51, 49, 46, 30); both colourings are in the files `H231_5colouring.txt` and `H961_5colouring.txt` and were checked edge by edge. Hence  $\chi = 5$ .

**(iv) Criticality and spindles.** For each vertex  $v$  the files `H231_criticality_certificates.txt` and `H961_criticality_certificates.txt` contain a proper 4-colouring of the graph with  $v$  deleted (231 and 961 certificates); each was checked edge by edge. Hence both graphs are vertex-critical. An exhaustive search over all pairs of rhombi (pairs of triangles sharing an edge) with a common apex whose tips are adjacent finds no Moser spindle in either graph.  $\square$

## 5 How the graphs were found

**Encoding with selectors.** To test many vertex subsets  $U \subseteq V$  we add a selector literal  $s_v$  to every at-least-one-colour clause,  $\neg s_v \vee \bigvee_c x_{v,c}$ , and solve under the assumptions  $\{s_v : v \in U\}$ , with the colours of one triangle of  $U$  fixed. An unsatisfiable answer comes with a *core*: a subset of the assumptions that is already unsatisfiable, i.e. a non-4-colourable subset of  $U$ .

**Algorithm 1 (random-order vertex deletion).**

1. Set  $U \leftarrow V$  and  $C \leftarrow \emptyset$  (vertices known to be essential).
2. Repeatedly remove vertices of degree  $< 4$  in  $G[U]$  (they never affect 4-colourability).
3. Pick a random  $v \in U \setminus C$ . If  $G[U \setminus \{v\}]$  is not 4-colourable, replace  $U$  by the returned core (and go to step 2); otherwise add  $v$  to  $C$ .
4. Stop when  $C = U$ .

Each test was run with a conflict limit and an inconclusive test was treated as “4-colourable”, so the output is vertex-critical only up to these limits; criticality of the final graphs  $H_{231}$  and  $H_{961}$  was verified separately (Section 4). For  $G_{372}$  each test takes well under a second. Sixteen runs with different random orders gave non-4-colourable subgraphs with between 235 and 337 vertices (Figure 2).

**Algorithm 2 (local search).** Starting from a vertex-critical  $U$ : add to  $U$  between 5 and 40 random vertices of  $V \setminus U$  having at least 2 to 4 neighbours in  $U$ , optionally delete a few random vertices of  $U$ , run Algorithm 1 on the resulting set, and accept the result if it is not larger than  $U$ . Starting from the 235-vertex subgraph this produced the 231-vertex graph  $H_{231}$ .

**Triage for  $G_{972}$ .** For  $G_{972}$  each test takes several seconds, so we first tested every vertex  $v$  of a 5-chromatic subgraph  $U$  with  $|U| = 968$  separately. If  $G[U \setminus \{v\}]$  is 4-colourable, then  $v$  lies in every non-4-colourable subgraph of  $G[U]$  and need never be tested again. Only 34 of the 968 vertices were individually removable (931 were shown to be essential; the three vertices of the triangle whose colours were fixed were not tested); deleting the 34 greedily gave 963 vertices, and Algorithm 2 gave 961.

*Remark 3.* Unlike  $G_{372}$ , the graph  $G_{972}$  is close to vertex-critical: at least 931 of the 968 vertices of the first subgraph examined are essential.

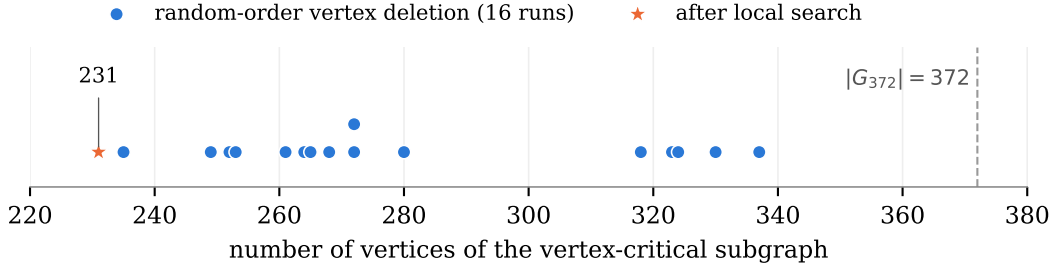


Figure 2: Sizes of the non-4-colourable subgraphs of  $G_{372}$  found by 16 runs of Algorithm 1 with different random orders, and the result of Algorithm 2.

## 6 Data availability

All data, proofs and code are in the repository accompanying this paper ([repository DOI]): for both graphs the vertex lists (parent-graph indices and 30-digit coordinates), exact coordinates in nested radicals (Mathematica and Python syntax), edge lists, 5-colourings, criticality certificates, CNF formulas with compressed DRAT proofs and DRAT-TRIM logs, the scripts that rebuild  $G_{372}$  and  $G_{972}$ , and the verification programs. A single script (`verify_all.sh`) re-runs every check of Section 4. The Supplementary Material describes the files and procedures and lists all coordinates in radicals.

## 7 Concluding remarks

The graph  $H_{231}$  is a local optimum of our search; determining the smallest 5-chromatic subgraph of  $G_{372}$ , or the smallest 5-chromatic unit-distance graph in  $S^2(r_1)$ , remains open. An exact implicit-hitting-set computation of the former did not finish within the time we allotted. The other constructions of [10] may also contain much smaller 5-chromatic subgraphs.

## Use of AI tools

Generative-AI tools were used extensively for computational exploration, for implementing the search and verification scripts, and for drafting the text. The author chose the research direction, set the acceptance and verification criteria, reviewed the results, decided which constructions and claims to include, and takes full responsibility for all claims. Every claim rests on certificates or exact computations that can be re-checked independently (Section 4).

## References

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## A Vertex list and a 5-colouring of $H_{231}$

Each row gives an index, the orbit of  $G_{372}$  (I: icosahedron vertices; 1, 2, 3: orbits of  $v_1, v_2, v_3$ ), the coordinates rounded to four decimals (labels of exact vertices of  $G_{372}$ , see Section 3), and a colour 1–5 of a proper 5-colouring.

| #  | orb. | $x$     | $y$     | $z$     | col. |
|----|------|---------|---------|---------|------|
| 1  | I    | 0.8090  | 0.5000  | 0.0000  | 1    |
| 2  | I    | −0.8090 | −0.5000 | 0.0000  | 5    |
| 3  | I    | 0.0000  | 0.8090  | 0.5000  | 4    |
| 4  | I    | 0.8090  | −0.5000 | 0.0000  | 3    |
| 5  | I    | 0.0000  | −0.8090 | −0.5000 | 1    |
| 6  | I    | −0.8090 | 0.5000  | 0.0000  | 1    |
| 7  | I    | 0.0000  | −0.8090 | 0.5000  | 4    |
| 8  | I    | 0.5000  | 0.0000  | 0.8090  | 2    |
| 9  | I    | −0.5000 | −0.0000 | 0.8090  | 3    |
| 10 | I    | 0.0000  | 0.8090  | −0.5000 | 3    |
| 11 | I    | −0.5000 | 0.0000  | −0.8090 | 4    |
| 12 | I    | 0.5000  | 0.0000  | −0.8090 | 2    |
| 13 | 1    | 0.7158  | 0.3492  | 0.5197  | 2    |
| 14 | 1    | −0.7158 | −0.3492 | 0.5197  | 5    |
| 15 | 1    | 0.5197  | 0.7158  | 0.3492  | 2    |
| 16 | 1    | 0.2360  | 0.4272  | 0.8163  | 2    |
| 17 | 1    | 0.5197  | −0.7158 | −0.3492 | 3    |
| 18 | 1    | −0.5197 | −0.7158 | 0.3492  | 3    |
| 19 | 1    | 0.3492  | 0.5197  | 0.7158  | 2    |
| 20 | 1    | −0.2113 | 0.4671  | 0.8011  | 2    |
| 21 | 1    | 0.5197  | −0.7158 | 0.3492  | 2    |
| 22 | 1    | 0.8163  | 0.2360  | 0.4272  | 3    |
| 23 | 1    | 0.0246  | −0.0852 | 0.9469  | 1    |

| #   | orb. | $x$     | $y$     | $z$     | col. |
|-----|------|---------|---------|---------|------|
| 59  | 1    | 0.7311  | -0.3739 | -0.4799 | 5    |
| 60  | 1    | -0.8011 | 0.2113  | 0.4671  | 2    |
| 61  | 1    | -0.0852 | -0.9469 | 0.0246  | 2    |
| 62  | 1    | -0.0246 | -0.0852 | -0.9469 | 5    |
| 63  | 1    | -0.9469 | 0.0246  | 0.0852  | 2    |
| 64  | 1    | 0.0246  | -0.0852 | -0.9469 | 2    |
| 65  | 1    | -0.4272 | 0.8163  | -0.2360 | 1    |
| 66  | 1    | 0.4671  | 0.8011  | -0.2113 | 5    |
| 67  | 1    | 0.4799  | 0.7311  | 0.3739  | 4    |
| 68  | 1    | 0.3739  | -0.4799 | -0.7311 | 5    |
| 69  | 1    | 0.4671  | -0.8011 | -0.2113 | 5    |
| 70  | 1    | -0.7158 | 0.3492  | -0.5197 | 3    |
| 71  | 1    | 0.3739  | 0.4799  | -0.7311 | 2    |
| 72  | 1    | -0.2360 | 0.4272  | -0.8163 | 2    |
| 73  | 1    | 0.8011  | 0.2113  | -0.4671 | 1    |
| 74  | 1    | 0.7158  | -0.3492 | -0.5197 | 5    |
| 75  | 1    | 0.4671  | -0.8011 | 0.2113  | 4    |
| 76  | 1    | 0.9469  | -0.0246 | 0.0852  | 4    |
| 77  | 1    | -0.0246 | -0.0852 | 0.9469  | 1    |
| 78  | 1    | 0.7158  | 0.3492  | -0.5197 | 1    |
| 79  | 1    | 0.7311  | -0.3739 | 0.4799  | 3    |
| 80  | 1    | 0.2113  | 0.4671  | 0.8011  | 3    |
| 81  | 1    | 0.8011  | -0.2113 | -0.4671 | 5    |
| 82  | 1    | -0.4671 | 0.8011  | -0.2113 | 3    |
| 83  | 1    | -0.9469 | -0.0246 | 0.0852  | 2    |
| 84  | 1    | -0.8163 | 0.2360  | 0.4272  | 5    |
| 85  | 1    | -0.3739 | 0.4799  | 0.7311  | 1    |
| 86  | 1    | 0.4272  | -0.8163 | 0.2360  | 3    |
| 87  | 1    | 0.7311  | 0.3739  | 0.4799  | 4    |
| 88  | 1    | 0.9469  | -0.0246 | -0.0852 | 1    |
| 89  | 1    | -0.8163 | -0.2360 | -0.4272 | 5    |
| 90  | 1    | 0.2113  | -0.4671 | -0.8011 | 4    |
| 91  | 1    | -0.3739 | -0.4799 | 0.7311  | 5    |
| 92  | 1    | -0.4671 | -0.8011 | 0.2113  | 4    |
| 93  | 1    | -0.4799 | -0.7311 | -0.3739 | 1    |
| 94  | 1    | -0.9469 | 0.0246  | -0.0852 | 2    |
| 95  | 1    | -0.7158 | -0.3492 | -0.5197 | 3    |
| 96  | 1    | -0.2113 | -0.4671 | -0.8011 | 1    |
| 97  | 1    | -0.7311 | 0.3739  | -0.4799 | 1    |
| 98  | 1    | -0.4799 | 0.7311  | -0.3739 | 1    |
| 99  | 1    | -0.0246 | 0.0852  | -0.9469 | 4    |
| 100 | 1    | -0.4671 | -0.8011 | -0.2113 | 1    |
| 101 | 1    | -0.3739 | 0.4799  | -0.7311 | 3    |
| 102 | 1    | -0.8011 | -0.2113 | -0.4671 | 2    |
| 103 | 1    | -0.3739 | -0.4799 | -0.7311 | 1    |
| 104 | 1    | -0.2360 | -0.4272 | -0.8163 | 3    |
| 105 | 1    | -0.4671 | 0.8011  | 0.2113  | 1    |
| 106 | 1    | 0.0852  | 0.9469  | -0.0246 | 3    |
| 107 | 1    | 0.4799  | -0.7311 | -0.3739 | 2    |
| 108 | 1    | 0.0852  | -0.9469 | -0.0246 | 2    |
| 109 | 1    | 0.4799  | 0.7311  | -0.3739 | 1    |
| 110 | 1    | 0.2113  | 0.4671  | -0.8011 | 2    |
| 111 | 1    | -0.8011 | 0.2113  | -0.4671 | 4    |
| 112 | 2    | -0.0796 | -0.0834 | 0.9440  | 5    |
| 113 | 2    | 0.0796  | 0.0834  | 0.9440  | 1    |
| 114 | 2    | 0.9440  | 0.0796  | 0.0834  | 1    |
| 115 | 2    | -0.0834 | 0.9440  | -0.0796 | 5    |
| 116 | 2    | 0.9440  | 0.0796  | -0.0834 | 1    |

| #   | orb. | $x$     | $y$     | $z$     | col. |
|-----|------|---------|---------|---------|------|
| 117 | 2    | 0.8301  | -0.2640 | 0.3818  | 3    |
| 118 | 2    | 0.3194  | 0.5622  | 0.6974  | 5    |
| 119 | 2    | 0.1844  | -0.5106 | 0.7809  | 2    |
| 120 | 2    | -0.9440 | -0.0796 | 0.0834  | 2    |
| 121 | 2    | 0.0834  | 0.9440  | 0.0796  | 2    |
| 122 | 2    | -0.2640 | 0.3818  | 0.8301  | 4    |
| 123 | 2    | 0.6974  | -0.3194 | 0.5622  | 5    |
| 124 | 2    | 0.3990  | -0.4334 | 0.7466  | 2    |
| 125 | 2    | -0.5622 | -0.6974 | -0.3194 | 5    |
| 126 | 2    | -0.9440 | -0.0796 | -0.0834 | 2    |
| 127 | 2    | -0.8301 | 0.2640  | 0.3818  | 1    |
| 128 | 2    | -0.0834 | -0.9440 | -0.0796 | 1    |
| 129 | 2    | 0.1844  | 0.5106  | 0.7809  | 3    |
| 130 | 2    | 0.5106  | -0.7809 | 0.1844  | 4    |
| 131 | 2    | 0.3818  | 0.8301  | 0.2640  | 4    |
| 132 | 2    | -0.3194 | -0.5622 | 0.6974  | 1    |
| 133 | 2    | -0.6974 | -0.3194 | -0.5622 | 4    |
| 134 | 2    | 0.3818  | 0.8301  | -0.2640 | 4    |
| 135 | 2    | 0.6974  | 0.3194  | 0.5622  | 5    |
| 136 | 2    | 0.5622  | 0.6974  | 0.3194  | 5    |
| 137 | 2    | 0.7809  | 0.1844  | -0.5106 | 1    |
| 138 | 2    | -0.3818 | -0.8301 | -0.2640 | 3    |
| 139 | 2    | 0.0834  | -0.9440 | 0.0796  | 3    |
| 140 | 2    | 0.4334  | -0.7466 | 0.3990  | 3    |
| 141 | 2    | -0.2640 | -0.3818 | 0.8301  | 4    |
| 142 | 2    | 0.8301  | -0.2640 | -0.3818 | 5    |
| 143 | 2    | 0.5622  | 0.6974  | -0.3194 | 3    |
| 144 | 2    | 0.7809  | 0.1844  | 0.5106  | 1    |
| 145 | 2    | -0.3990 | 0.4334  | 0.7466  | 3    |
| 146 | 2    | 0.7466  | 0.3990  | -0.4334 | 4    |
| 147 | 2    | 0.7466  | -0.3990 | -0.4334 | 4    |
| 148 | 2    | -0.3194 | -0.5622 | -0.6974 | 1    |
| 149 | 2    | -0.0796 | -0.0834 | -0.9440 | 4    |
| 150 | 2    | 0.4334  | -0.7466 | -0.3990 | 2    |
| 151 | 2    | -0.5106 | 0.7809  | 0.1844  | 1    |
| 152 | 2    | -0.3818 | -0.8301 | 0.2640  | 4    |
| 153 | 2    | -0.6974 | 0.3194  | -0.5622 | 4    |
| 154 | 2    | -0.7466 | -0.3990 | 0.4334  | 3    |
| 155 | 2    | -0.5622 | -0.6974 | 0.3194  | 2    |
| 156 | 2    | -0.7466 | 0.3990  | 0.4334  | 1    |
| 157 | 2    | -0.7809 | -0.1844 | -0.5106 | 3    |
| 158 | 2    | 0.3990  | -0.4334 | -0.7466 | 2    |
| 159 | 2    | 0.0796  | 0.0834  | -0.9440 | 4    |
| 160 | 2    | 0.5106  | -0.7809 | -0.1844 | 3    |
| 161 | 2    | -0.4334 | 0.7466  | 0.3990  | 4    |
| 162 | 2    | -0.8301 | 0.2640  | -0.3818 | 1    |
| 163 | 2    | -0.7809 | -0.1844 | 0.5106  | 2    |
| 164 | 2    | -0.1844 | 0.5106  | -0.7809 | 3    |
| 165 | 2    | -0.4334 | 0.7466  | -0.3990 | 5    |
| 166 | 2    | 0.3194  | 0.5622  | -0.6974 | 3    |
| 167 | 2    | 0.2640  | 0.3818  | -0.8301 | 2    |
| 168 | 2    | 0.2640  | -0.3818 | -0.8301 | 5    |
| 169 | 2    | -0.3990 | 0.4334  | -0.7466 | 3    |
| 170 | 2    | -0.5106 | 0.7809  | -0.1844 | 1    |
| 171 | 2    | -0.1844 | -0.5106 | -0.7809 | 1    |
| 172 | 3    | -0.1765 | 0.7993  | 0.4843  | 1    |
| 173 | 3    | 0.1765  | -0.7993 | 0.4843  | 4    |
| 174 | 3    | 0.4843  | -0.1765 | 0.7993  | 4    |

| #   | orb. | $x$     | $y$     | $z$     | col. |
|-----|------|---------|---------|---------|------|
| 175 | 3    | -0.4843 | 0.1765  | 0.7993  | 4    |
| 176 | 3    | 0.7993  | 0.4843  | -0.1765 | 1    |
| 177 | 3    | 0.5852  | 0.1379  | 0.7369  | 2    |
| 178 | 3    | 0.0467  | -0.8845 | 0.3464  | 4    |
| 179 | 3    | 0.4843  | -0.1765 | -0.7993 | 1    |
| 180 | 3    | 0.7081  | -0.6319 | -0.0624 | 3    |
| 181 | 3    | -0.7993 | 0.4843  | 0.1765  | 3    |
| 182 | 3    | 0.7369  | 0.5852  | 0.1379  | 2    |
| 183 | 3    | 0.8845  | 0.3464  | 0.0467  | 1    |
| 184 | 3    | -0.4088 | 0.1476  | 0.8459  | 3    |
| 185 | 3    | -0.1379 | -0.7369 | 0.5852  | 1    |
| 186 | 3    | 0.7993  | -0.4843 | -0.1765 | 3    |
| 187 | 3    | 0.6319  | 0.0624  | 0.7081  | 2    |
| 188 | 3    | 0.3464  | 0.0467  | 0.8845  | 3    |
| 189 | 3    | 0.5852  | -0.1379 | 0.7369  | 2    |
| 190 | 3    | 0.7369  | 0.5852  | -0.1379 | 1    |
| 191 | 3    | 0.3464  | 0.0467  | -0.8845 | 3    |
| 192 | 3    | 0.8459  | -0.4088 | -0.1476 | 5    |
| 193 | 3    | 0.0467  | 0.8845  | 0.3464  | 4    |
| 194 | 3    | -0.4843 | 0.1765  | -0.7993 | 1    |
| 195 | 3    | 0.1379  | 0.7369  | -0.5852 | 2    |
| 196 | 3    | -0.7081 | 0.6319  | -0.0624 | 1    |
| 197 | 3    | -0.0624 | 0.7081  | -0.6319 | 3    |
| 198 | 3    | -0.3464 | -0.0467 | -0.8845 | 1    |
| 199 | 3    | -0.7369 | -0.5852 | 0.1379  | 2    |
| 200 | 3    | -0.7993 | -0.4843 | 0.1765  | 2    |
| 201 | 3    | -0.4088 | -0.1476 | 0.8459  | 1    |
| 202 | 3    | -0.1476 | -0.8459 | -0.4088 | 2    |
| 203 | 3    | -0.8845 | -0.3464 | 0.0467  | 3    |
| 204 | 3    | -0.0467 | -0.8845 | -0.3464 | 1    |
| 205 | 3    | 0.1379  | 0.7369  | 0.5852  | 2    |
| 206 | 3    | 0.8459  | -0.4088 | 0.1476  | 1    |
| 207 | 3    | 0.7081  | -0.6319 | 0.0624  | 3    |
| 208 | 3    | -0.0624 | -0.7081 | -0.6319 | 2    |
| 209 | 3    | -0.1765 | 0.7993  | -0.4843 | 2    |
| 210 | 3    | -0.1476 | -0.8459 | 0.4088  | 1    |
| 211 | 3    | -0.6319 | -0.0624 | 0.7081  | 2    |
| 212 | 3    | 0.0624  | 0.7081  | 0.6319  | 4    |
| 213 | 3    | -0.3464 | -0.0467 | 0.8845  | 3    |
| 214 | 3    | -0.8845 | -0.3464 | -0.0467 | 2    |
| 215 | 3    | -0.7369 | -0.5852 | -0.1379 | 3    |
| 216 | 3    | -0.8459 | 0.4088  | -0.1476 | 4    |
| 217 | 3    | -0.1379 | -0.7369 | -0.5852 | 4    |
| 218 | 3    | 0.1765  | -0.7993 | -0.4843 | 2    |
| 219 | 3    | 0.6319  | 0.0624  | -0.7081 | 4    |
| 220 | 3    | 0.1476  | 0.8459  | -0.4088 | 3    |
| 221 | 3    | -0.0467 | 0.8845  | -0.3464 | 2    |
| 222 | 3    | 0.0624  | -0.7081 | 0.6319  | 4    |
| 223 | 3    | -0.8459 | 0.4088  | 0.1476  | 1    |
| 224 | 3    | -0.7081 | 0.6319  | 0.0624  | 4    |
| 225 | 3    | 0.1476  | 0.8459  | 0.4088  | 3    |
| 226 | 3    | -0.5852 | 0.1379  | -0.7369 | 1    |
| 227 | 3    | 0.8845  | 0.3464  | -0.0467 | 4    |
| 228 | 3    | 0.4088  | 0.1476  | -0.8459 | 4    |
| 229 | 3    | 0.4088  | -0.1476 | -0.8459 | 2    |
| 230 | 3    | -0.6319 | -0.0624 | -0.7081 | 4    |
| 231 | 3    | -0.5852 | -0.1379 | -0.7369 | 4    |

## B Exact orbit representatives

With  $b = \sqrt{\beta}$  and  $g = \sqrt{\gamma}$  as in Lemma 2 (separately for each sphere, all square roots positive), the orbit representatives of  $G_{372}$  and  $G_{972}$  are as follows. Every vertex of  $H_{231}$  (resp.  $H_{961}$ ) is  $Mv_i$  (resp.  $M'w_i$ ) for a matrix  $M \in \Gamma$  (resp.  $M' \in \Gamma'$ ), or a vertex of the (great) icosahedron;

together with the generators of  $\Gamma$  in Section 2 this defines every vertex exactly within this paper. The four-decimal labels of Appendix A and Table 2 select the vertices.

$v_1$  (on  $S^2(r_1)$ )

$$\begin{aligned}x &= \frac{1}{4} - \frac{1}{8}b + \frac{1}{8}\sqrt{5}b \\y &= -\frac{1}{8} - \frac{1}{8}\sqrt{5} + \frac{1}{4}b \\z &= \frac{1}{4}g + \frac{1}{4}\sqrt{5}g\end{aligned}$$

$v_2$  (on  $S^2(r_1)$ )

$$\begin{aligned}x &= \frac{1}{16} + \frac{3}{16}\sqrt{5} - \frac{1}{16}b - \frac{1}{16}\sqrt{5}b + \frac{5}{16}g + \frac{1}{16}\sqrt{5}g - \frac{1}{8}bg \\y &= \frac{1}{8} - \frac{1}{16}b + \frac{1}{16}\sqrt{5}b + \frac{1}{16}g - \frac{1}{16}\sqrt{5}g - \frac{1}{16}bg - \frac{1}{16}\sqrt{5}bg \\z &= \frac{3}{16} + \frac{3}{16}\sqrt{5} - \frac{1}{8}b + \frac{1}{8}g + \frac{3}{16}bg + \frac{1}{16}\sqrt{5}bg\end{aligned}$$

$v_3$  (on  $S^2(r_1)$ )

$$\begin{aligned}x &= \frac{1}{16} + \frac{3}{16}\sqrt{5} - \frac{1}{16}b - \frac{1}{16}\sqrt{5}b - \frac{5}{16}g - \frac{1}{16}\sqrt{5}g + \frac{1}{8}bg \\y &= \frac{1}{8} - \frac{1}{16}b + \frac{1}{16}\sqrt{5}b - \frac{1}{16}g + \frac{1}{16}\sqrt{5}g + \frac{1}{16}bg + \frac{1}{16}\sqrt{5}bg \\z &= -\frac{3}{16} - \frac{3}{16}\sqrt{5} + \frac{1}{8}b + \frac{1}{8}g + \frac{3}{16}bg + \frac{1}{16}\sqrt{5}bg\end{aligned}$$

$w_1$  (on  $S^2(r_2)$ )

$$\begin{aligned}x &= -\frac{5}{8} + \frac{1}{8}\sqrt{5} + \frac{3}{8}b - \frac{1}{8}\sqrt{5}b - \frac{3}{16}g + \frac{1}{16}\sqrt{5}g - \frac{1}{16}bg + \frac{1}{16}\sqrt{5}bg \\y &= -\frac{1}{8} - \frac{1}{8}\sqrt{5} - \frac{1}{2}b + \frac{1}{4}\sqrt{5}b - \frac{1}{4}g + \frac{1}{8}\sqrt{5}g - \frac{3}{16}bg + \frac{1}{16}\sqrt{5}bg \\z &= -\frac{5}{8} + \frac{1}{8}\sqrt{5} + \frac{3}{16}g + \frac{1}{16}\sqrt{5}g + \frac{1}{4}bg - \frac{1}{8}\sqrt{5}bg\end{aligned}$$

$w_2$  (on  $S^2(r_2)$ )

$$\begin{aligned}x &= -\frac{5}{8} + \frac{1}{8}\sqrt{5} + \frac{3}{8}b - \frac{1}{8}\sqrt{5}b + \frac{3}{16}g - \frac{1}{16}\sqrt{5}g + \frac{1}{16}bg - \frac{1}{16}\sqrt{5}bg \\y &= -\frac{1}{8} - \frac{1}{8}\sqrt{5} - \frac{1}{2}b + \frac{1}{4}\sqrt{5}b + \frac{1}{4}g - \frac{1}{8}\sqrt{5}g + \frac{3}{16}bg - \frac{1}{16}\sqrt{5}bg \\z &= \frac{5}{8} - \frac{1}{8}\sqrt{5} + \frac{3}{16}g + \frac{1}{16}\sqrt{5}g + \frac{1}{4}bg - \frac{1}{8}\sqrt{5}bg\end{aligned}$$

$w_3$  (on  $S^2(r_2)$ )

$$\begin{aligned}x &= -\frac{159}{656} - \frac{69}{656}\sqrt{5} - \frac{23}{328}b + \frac{7}{164}\sqrt{5}b + \frac{3}{164}g - \frac{9}{328}\sqrt{5}g + \frac{29}{656}bg - \frac{23}{656}\sqrt{5}bg \\y &= -\frac{257}{656} + \frac{119}{656}\sqrt{5} + \frac{93}{656}b - \frac{37}{656}\sqrt{5}b - \frac{51}{656}g + \frac{15}{656}\sqrt{5}g - \frac{9}{82}bg + \frac{13}{328}\sqrt{5}bg \\z &= -\frac{87}{328} + \frac{7}{82}\sqrt{5} - \frac{113}{656}b + \frac{67}{656}\sqrt{5}b + \frac{17}{656}g - \frac{87}{656}\sqrt{5}g - \frac{17}{656}bg + \frac{5}{656}\sqrt{5}bg\end{aligned}$$

$w_4$  (on  $S^2(r_2)$ )

$$\begin{aligned}x &= -\frac{159}{656} - \frac{69}{656}\sqrt{5} - \frac{23}{328}b + \frac{7}{164}\sqrt{5}b - \frac{3}{164}g + \frac{9}{328}\sqrt{5}g - \frac{29}{656}bg + \frac{23}{656}\sqrt{5}bg \\y &= \frac{257}{656} - \frac{119}{656}\sqrt{5} - \frac{93}{656}b + \frac{37}{656}\sqrt{5}b - \frac{51}{656}g + \frac{15}{656}\sqrt{5}g - \frac{9}{82}bg + \frac{13}{328}\sqrt{5}bg \\z &= \frac{87}{328} - \frac{7}{82}\sqrt{5} + \frac{113}{656}b - \frac{67}{656}\sqrt{5}b + \frac{17}{656}g - \frac{87}{656}\sqrt{5}g - \frac{17}{656}bg + \frac{5}{656}\sqrt{5}bg\end{aligned}$$

$w_5$  (on  $S^2(r_2)$ )

$$\begin{aligned} x &= -\frac{169}{656} + \frac{69}{656}\sqrt{5} + \frac{23}{328}b - \frac{7}{164}\sqrt{5}b - \frac{3}{164}g + \frac{9}{328}\sqrt{5}g - \frac{29}{656}bg + \frac{23}{656}\sqrt{5}bg \\ y &= \frac{93}{656} + \frac{45}{656}\sqrt{5} - \frac{93}{656}b + \frac{37}{656}\sqrt{5}b + \frac{51}{656}g - \frac{15}{656}\sqrt{5}g + \frac{9}{82}bg - \frac{13}{328}\sqrt{5}bg \\ z &= -\frac{9}{82} + \frac{13}{328}\sqrt{5} + \frac{31}{656}b + \frac{15}{656}\sqrt{5}b + \frac{147}{656}g - \frac{77}{656}\sqrt{5}g + \frac{17}{656}bg - \frac{5}{656}\sqrt{5}bg \end{aligned}$$

$w_6$  (on  $S^2(r_2)$ )

$$\begin{aligned} x &= -\frac{169}{656} + \frac{69}{656}\sqrt{5} + \frac{23}{328}b - \frac{7}{164}\sqrt{5}b + \frac{3}{164}g - \frac{9}{328}\sqrt{5}g + \frac{29}{656}bg - \frac{23}{656}\sqrt{5}bg \\ y &= \frac{93}{656} + \frac{45}{656}\sqrt{5} - \frac{93}{656}b + \frac{37}{656}\sqrt{5}b - \frac{51}{656}g + \frac{15}{656}\sqrt{5}g - \frac{9}{82}bg + \frac{13}{328}\sqrt{5}bg \\ z &= \frac{9}{82} - \frac{13}{328}\sqrt{5} - \frac{31}{656}b - \frac{15}{656}\sqrt{5}b + \frac{147}{656}g - \frac{77}{656}\sqrt{5}g + \frac{17}{656}bg - \frac{5}{656}\sqrt{5}bg \end{aligned}$$

$w_7$  (on  $S^2(r_2)$ )

$$\begin{aligned} x &= -\frac{3}{8} + \frac{3}{16}b - \frac{1}{16}\sqrt{5}b - \frac{3}{16}g + \frac{1}{16}\sqrt{5}g + \frac{1}{16}bg - \frac{1}{16}\sqrt{5}bg \\ y &= \frac{1}{16} - \frac{1}{16}\sqrt{5} + \frac{1}{4}b - \frac{1}{8}\sqrt{5}b - \frac{1}{4}g + \frac{1}{8}\sqrt{5}g + \frac{3}{16}bg - \frac{1}{16}\sqrt{5}bg \\ z &= -\frac{7}{16} + \frac{1}{16}\sqrt{5} - \frac{1}{16}b + \frac{1}{16}\sqrt{5}b - \frac{5}{16}g + \frac{1}{16}\sqrt{5}g - \frac{1}{4}bg + \frac{1}{8}\sqrt{5}bg \end{aligned}$$

$w_8$  (on  $S^2(r_2)$ )

$$\begin{aligned} x &= -\frac{3}{8} + \frac{1}{8}\sqrt{5} + \frac{3}{16}g - \frac{1}{16}\sqrt{5}g + \frac{1}{16}bg - \frac{1}{16}\sqrt{5}bg \\ y &= \frac{1}{4} - \frac{1}{4}g + \frac{1}{8}\sqrt{5}g - \frac{3}{16}bg + \frac{1}{16}\sqrt{5}bg \\ z &= -\frac{1}{8} - \frac{1}{8}\sqrt{5} + \frac{3}{8}b - \frac{1}{8}\sqrt{5}b + \frac{5}{16}g - \frac{1}{16}\sqrt{5}g - \frac{1}{4}bg + \frac{1}{8}\sqrt{5}bg \end{aligned}$$